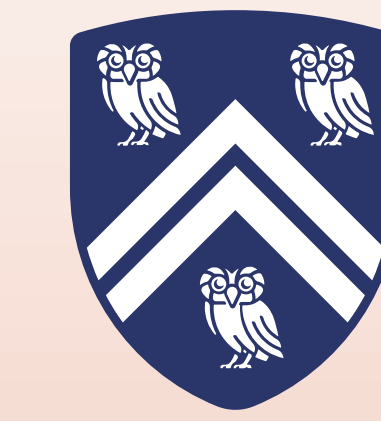




Learning User-Specific Sensory Strategies from Haptic Interaction

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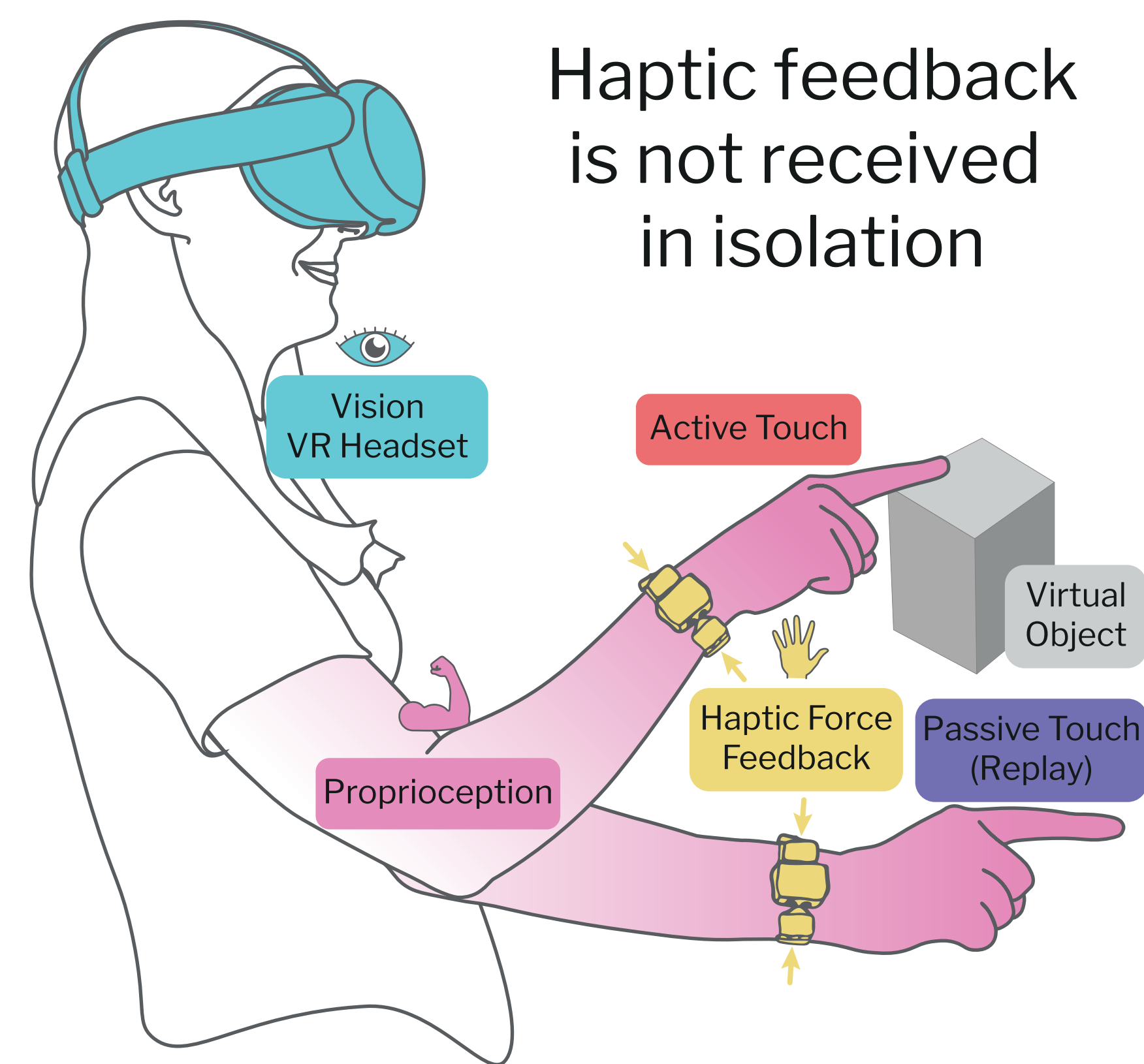


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People differ in how they combine visual, haptic, and proprioceptive information.
We infer those strategies directly from their interaction data.

Motivation



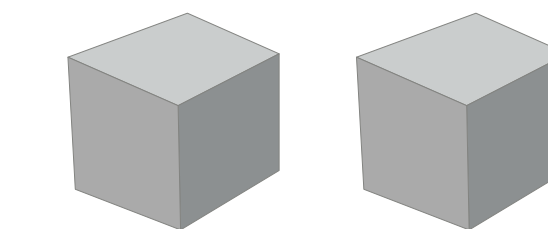
Two users can make the same perceptual decision while relying on different sensory evidence. Likewise, a user may reweight information when the interaction mode changes.

How Do We Infer Sensory Strategy?

Two VR Referred Haptics Studies [1, 2]

2-AFC virtual stiffness judgment
Wrist-worn squeeze feedback
Visual motion & proprioceptive movement
Study 1: Active Exploration (n = 16)
Study 2: Active Exploration & Passive Replay (n = 13)

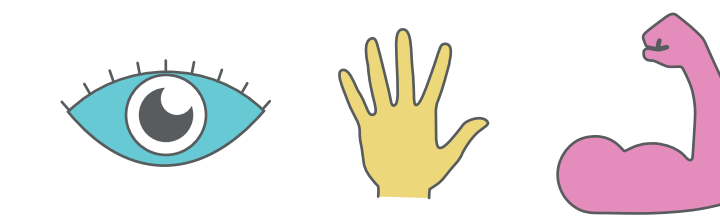
User's Perceptual Choice



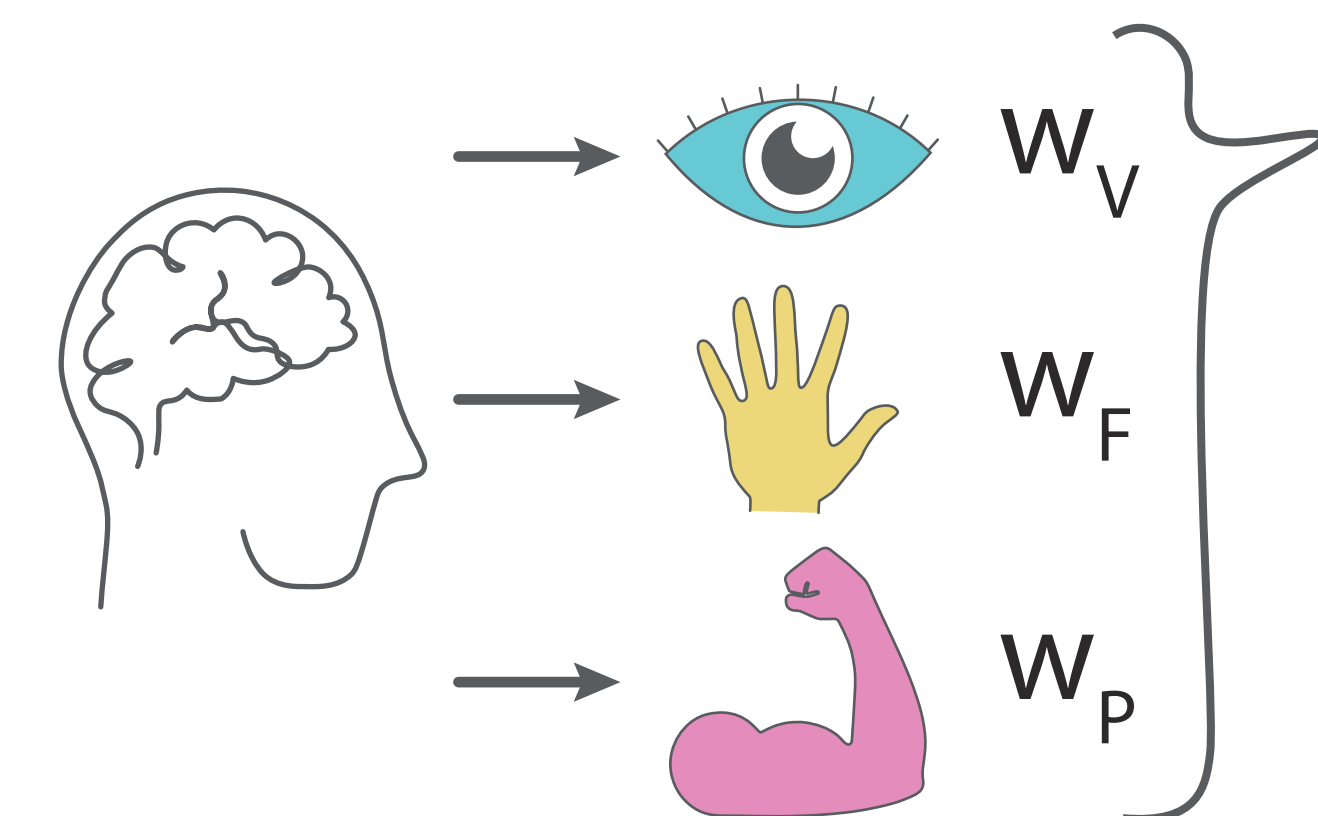
Which is stiffer?
(binary)

Recorded Interaction Evidence:

Visual motion, Haptic Force, Proprioceptive Movement,
Task Covariates (time, switches, experimental variables)



Hierarchical Bayesian Sensory Model



User Sensory Profile

$$\text{logit}(p_{it}) = \alpha_i + w_{F,i} x_{F,it} + w_{V,i} x_{V,it} + w_{P,i} x_{P,it} + \beta^T z_{it}$$

The weights quantify how strongly each interaction-derived sensory signal predicts a particular user's perceptual choices.

Hierarchical inference shares information across users while preserving individual differences.

(well suited to low-n, high-trial-count datasets)

Findings

Users Exhibit Distinct Sensory Strategies



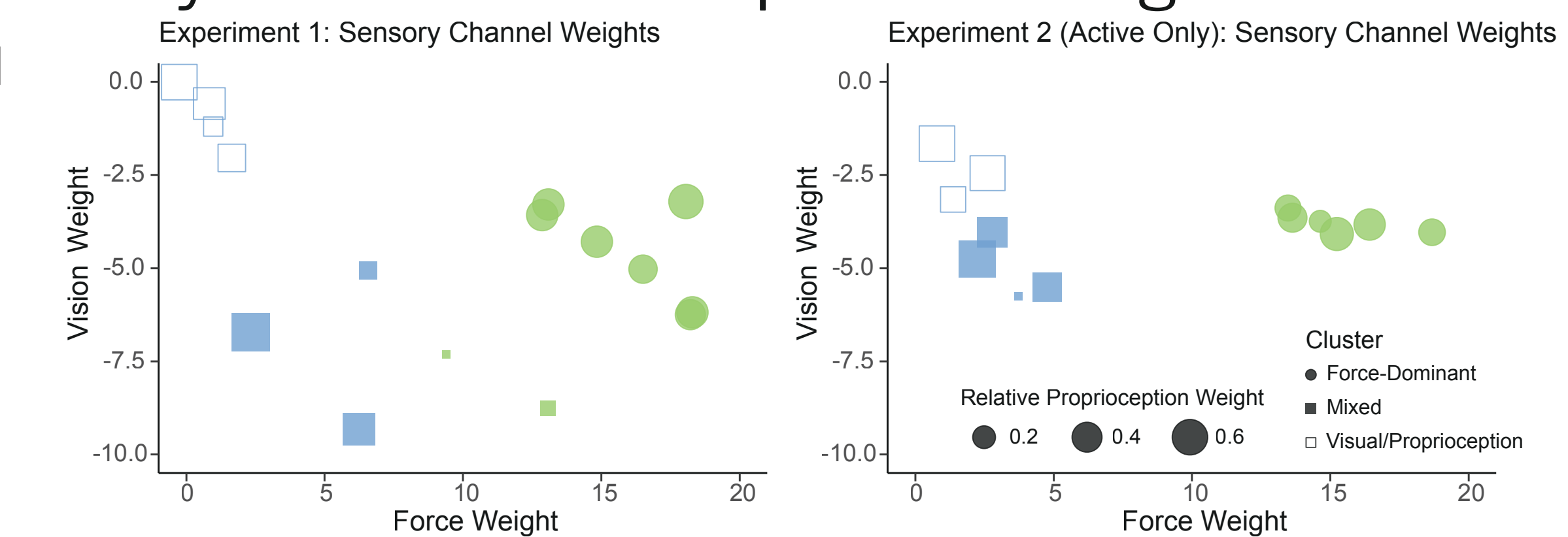
The same task produces different sensory-weight profiles across users.

Force-Dominant
-Mostly predicted by haptic force
Mixed
-Evidence distributed across modalities
Visual/Proprioception-Dominant
-Less reliance on haptic force

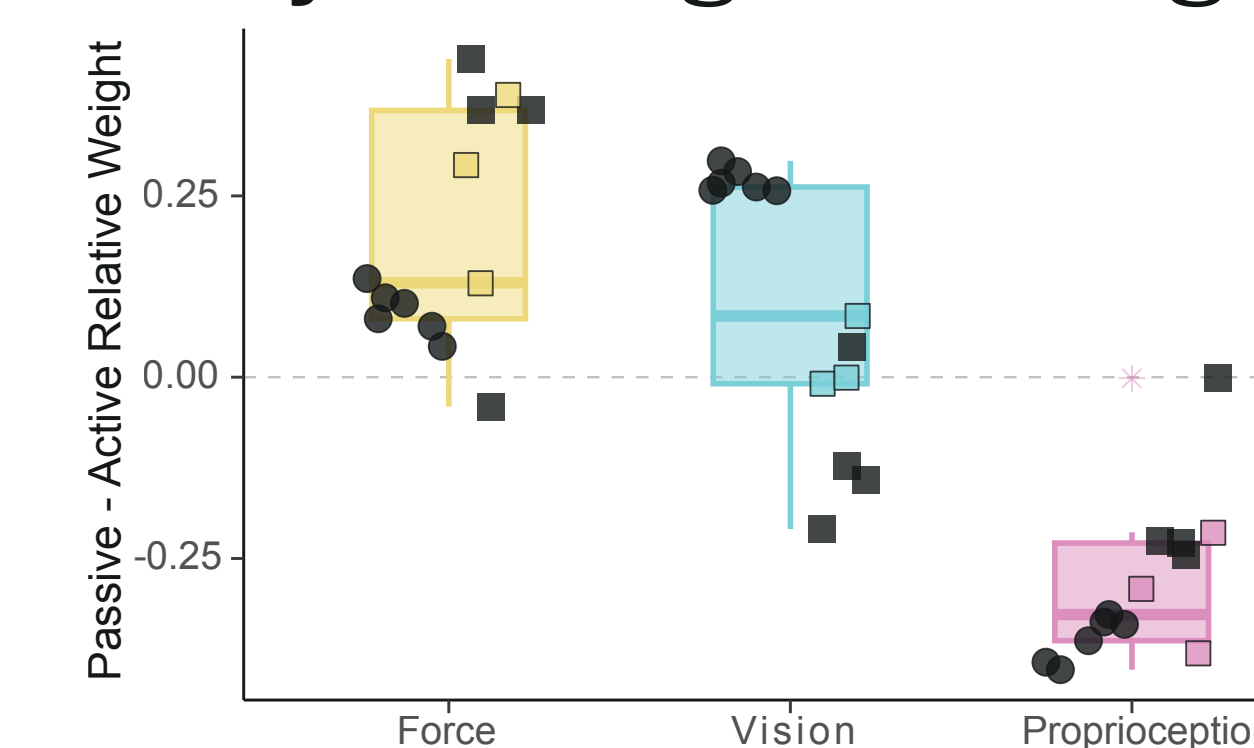
Model Recovers Independently Observed Perceptual Strategies

Prior study labels were not used for model fitting or clustering

Previously identified **Haptic-Prior** users were predominantly Force-dominant (Exp. 1: 7/9; Exp. 2: 6/6)
Previously identified **Visual-Prior** users split across the Mixed & Visual/Proprioception-dominant clusters



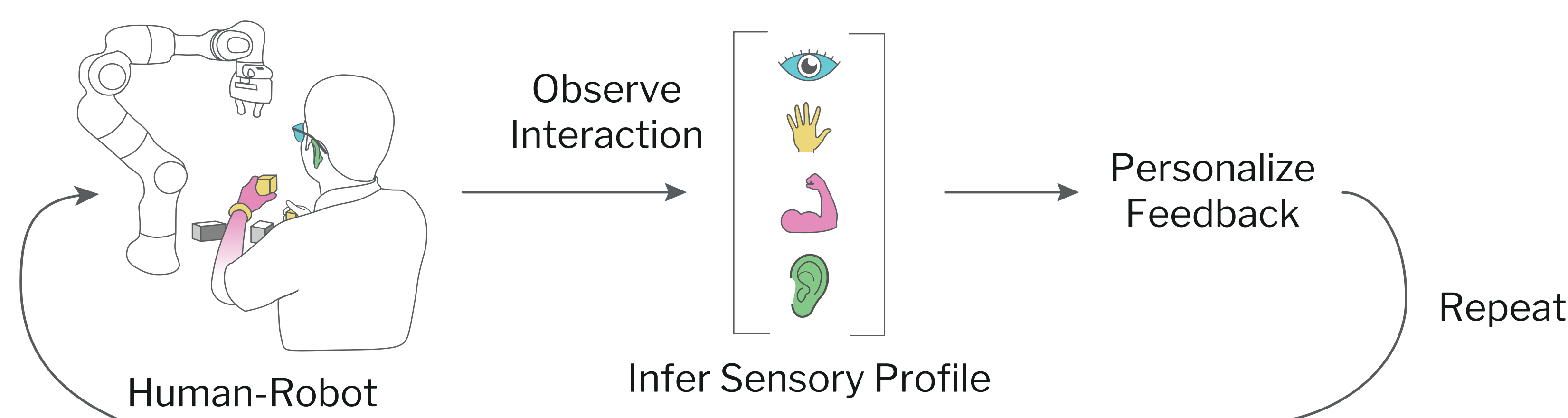
Sensory Strategies Change Without Active Control



Passive replay changes the relative composition of sensory evidence

Proprioceptive contribution generally decreases in passive replay, but the redistribution toward force and/or vision differs across users and profiles.

Toward Perceptually Adaptive Haptic Robotics



Some Future Applications:

»Teleoperation, Shared Control, Training & Assistance

Possible Questions to Address:

»Which cues should be emphasized?
»Does perception change when the robot takes control?
»Can feedback adapt as the user's strategy changes?

Key Takeaway: Haptic interaction data can reveal not only whether a user succeeds, but how that user interprets multisensory feedback. Modeling those individual sensory strategies is a step toward haptic systems that adapt feedback to the person and the interaction mode.

Sources

Under Review: Chase, E.D.Z., & O'Malley, M.K. (2026). Interaction-Informed Modeling Reveals User-Specific Sensory Weights in Active and Passive Haptics.

[1] Chase, E.D.Z., & O'Malley, M.K. (2024, June). The Interplay of Vision and Referred Haptic Feedback in VR Environments. In International Conference on Human Haptic Sensing and Touch Enabled Computer Applications (pp. 385-397). Cham: Springer Nature Switzerland.

[2] Chase, E.D.Z., Sullivan, D.H., & O'Malley, M.K. (2025, July). Hands-On or Hands-Off? Active Touch Influences Multisensory Perception of Referred Haptics. In 2025 IEEE World Haptics Conference (WHC) (pp. 321-331).